

MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)

1.Introduction

1.1Project Statement

This project aims to develop a comprehensive hospital operations and patient analytics dashboard suite for analyzing hospital performance, patient admissions, department efficiency, and healthcare resource utilization.

Using publicly available Hospital Datasets and Patient Admission Datasets, the project will transform healthcare operational data into actionable insights through interactive Tableau dashboards.

The dashboard suite will help hospital administrators, healthcare managers, medical staff, and policymakers monitor patient flow, evaluate department performance, optimize hospital resources, and improve operational efficiency through data-driven decision-making.

The final deliverable is a unified Tableau workbook consisting of four interconnected dashboards:

Hospital Overview

Patient Flow

Department Analytics

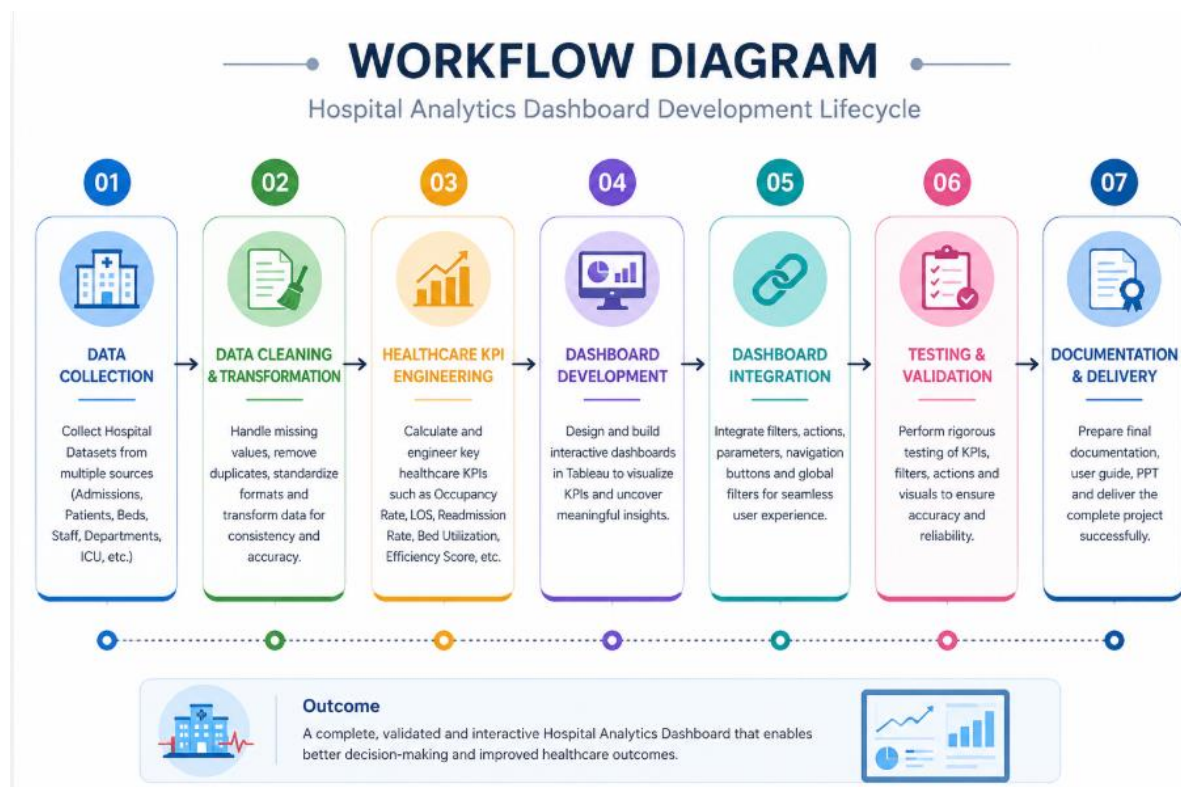
Resource Utilization

1.2 Objectives

The main objectives of this project are:

1. **To clean and preprocess hospital data** using Python by handling missing values, duplicates, inconsistent data, and incorrect data formats.
2. **To analyze hospital operations** and identify important patterns related to patients, admissions, departments, staff, beds, and other hospital resources.
3. **To develop meaningful Key Performance Indicators (KPIs)** such as Total Admissions, Occupancy Rate, Average Length of Stay, Readmission Rate, Bed Utilization Rate, and Department Efficiency Score.
4. **To create interactive Tableau dashboards** for Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization.
5. **To provide interactive data exploration** using filters, navigation controls, dashboard linking.
6. **To identify department-level and hospital-level insights** that can help in monitoring performance and resource utilization.
7. **To present complex hospital data through clear visualizations** that make the information easier to understand and support data-driven decision-making.
8. **To validate the dashboards and their functionality** through testing and a Quality Assurance (QA) checklist.

1.3 Workflow Diagram



2. Data Collection, Cleaning & Transformation

2.1 Dataset Overview

The project uses a **hospital healthcare dataset** containing patient-level information related to admissions, hospital departments, beds, staff, transfers, and patient outcomes. The dataset was used to develop healthcare KPIs and interactive Tableau dashboards for **Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization**.

The dataset was initially provided in Excel format and was processed using **Python/Pandas** before being used for dashboard development.

2.2 Data Source

The primary dataset used for the project was:

Dataset Name: Hospital_RawDataset_Updated

Working File: <https://www.kaggle.com/datasets/hassanjameelahmed/healthcare-analytics-patient-flow-data>

The dataset contains hospital and patient-related records that support analysis of patient flow, hospital capacity, departmental performance, and resource utilization.

2.3 Dataset Size

The initial dataset contained:

- **Rows:** 10,000 patient records

- **Columns:** Multiple patient, hospital, department, staff, bed, admission, discharge, and transfer-related attributes
- **Primary/Common Key:** Patient_ID

Each row represents a hospital-related patient record identified using Patient_ID.

2.4 Columns

The dataset contains features belonging to several analytical categories:

Category	Example Features	Purpose
Patient Information	Patient_ID, Patient-related attributes	Identifies and describes patients
Hospital Information	Hospital Name, Hospital Type, State, City	Enables hospital-level analysis
Admission Information	Admission Date, Admission Type	Analyzes patient admissions
Discharge Information	Discharge Date	Analyzes patient discharge and stay duration
Patient Flow	Transfer Date, Transfer-related information	Analyzes patient transfers
Department Information	Department	Enables department-wise analysis
Bed Information	Beds Occupied, ICU Beds	Measures hospital capacity and utilization
Staff Information	Staff Count	Measures staff availability and utilization
Length of Stay	Length of Stay	Measures duration of patient hospitalization

2.5. Data Cleaning and Transformation

2.5.1 Missing Values

Missing-value analysis was performed for all columns.

The most significant missing-value issue was identified in the Transfer_Date column. A large number of records did not contain a transfer date because the corresponding patients were not transferred.

Observed missing values in Transfer_Date: 8,240 records.

Since a missing transfer date could represent a patient who was never transferred, these values were handled by replacing them with:

Not Transferred

This makes the information easier to interpret during dashboard analysis rather than treating every missing value as an unknown value.

2.5.2 Duplicate Records

Duplicate records were checked using Pandas.

Duplicate records identified: 0

Therefore, no duplicate patient records required removal.

2.5.3 Incorrect Data Types

The data types of the columns were reviewed and corrected where necessary.

Particular attention was given to:

- Date columns
- Numerical columns
- Categorical/text columns
- Patient identifiers

Date fields were converted into appropriate datetime formats, while numerical measures were converted into suitable numeric data types.

2.5.4 Date Corrections

Date fields were converted into a standardized date format.

The following types of dates were examined:

- Admission Date
- Discharge Date
- Transfer Date

Date validation was also performed to identify logically invalid records.

2.5.5 Length-of-Stay Issues

The Length of Stay field was compared with the difference between admission and discharge dates.

A mismatch was identified in **6,051 records** between the existing Length of Stay value and the value derived from the admission and discharge dates.

These records were identified as part of the data-quality validation process and were reviewed before using Length of Stay for KPI calculations.

Additionally, **2,964 records** were identified where the discharge date occurred before the admission date. These records represent logically inconsistent date relationships and were flagged during the cleaning process.

This validation was important because incorrect Length of Stay values could directly affect the **Average Length of Stay** KPI and patient-flow analysis.

2.6 Data Quality Summary

Data Quality Check	Result	Action Taken
Initial Records	10,000	Retained for analysis
Duplicate Records	0	No removal required
Missing Transfer Dates	8,240	Replaced with Not Transferred
Length-of-Stay Mismatches	6,051	Identified and validated
Discharge Before Admission	2,964	Identified as invalid date relationships
Date Formatting	Inconsistent	Standardized
Data Types	Required correction	Converted to appropriate types
Categorical Values	Inconsistencies identified	Standardized
Final Dataset	Cleaned dataset	Used for KPI and dashboard development

3. KPI Engineering

3.1 Overview

Healthcare KPI Engineering is the process of converting the cleaned hospital dataset into meaningful performance indicators. These KPIs provide quantitative measures of hospital operations, patient flow, departmental performance, and resource utilization.

The calculated KPIs were designed to support the four project dashboards:

- Hospital Overview
- Patient Flow
- Department Analytics
- Resource Utilization

The KPIs were implemented using calculated fields and aggregation logic in Tableau.

3.2 KPI Definitions and Formulas

3.2.1 Total Admissions

Definition: Total number of patient admission records during the selected period.

Formula: Total Admissions = COUNT(Patient_ID)

Purpose: Measures the overall volume of patients admitted to the hospital.

Measurement : Count

3.2.2 Occupancy Rate

Definition: Percentage of available hospital beds occupied by patients.

Formula: $\text{Occupancy Rate} = \text{Avg}([\text{Bed Occupancy Rate Calc}])$

Purpose: Helps determine hospital capacity utilization.

Measurement : Percentage

3.2.3 Average Length of Stay

Definition: Average number of days patients remain hospitalized.

Formula: $\text{Average Length of Stay} = \text{AVG}(\text{DATEDIFF}('day', [\text{Admission Date}], [\text{Discharge Date}]))$

Purpose: Helps evaluate patient flow and hospital resource usage.

Measurement : Days

3.2.4 Readmission Rate

Definition: Percentage of patients who were readmitted to the hospital.

Formula: $\text{Readmission Rate} = \text{SUM}([\text{Readmission Flag}]) / \text{COUNT}([\text{Patient ID}])$

Purpose: Provides an indication of patient outcomes and potential follow-up care requirements.

Measurement : Percentage

3.2.5 Bed Utilization Rate

Definition: Percentage of available beds that are being utilized.

Formula: $\text{Bed Utilization Rate} = \text{SUM}([\text{Beds Occupied Count}]) / \text{SUM}([\text{Dept Bed Capacity Derived}]) * 100$

Purpose: Measures the utilization of hospital bed capacity.

Measurement : Percentage

3.2.6 Department Efficiency Score

Definition: A composite measure used to compare the operational performance of different hospital departments.

The score was derived using relevant departmental performance and resource-utilization measures.

Formula : $(\text{AVG}([\text{Staff Utilization \% Derived}]) + \text{AVG}([\text{Bed Occupancy Rate \%}])) / 2$

Purpose: Allows departments to be compared based on their operational efficiency.

Measurement : score

3.2.7 Staff Utilization Rate

Definition: Measures staff utilization relative to the number of occupied beds.

Formula: Staff Utilization Rate = ([Beds Occupied Count] / [Staff Count]) * 100

Purpose: Provides an indication of the relationship between patient load and available staff resources.

Measurement : Percentage

3.2.8 Equipment Availability Rate

Definition:

Percentage of required or available equipment that is currently available for use.

Formula: Equipment Availability Rate = COUNT(IF [Equipment Status] = "Available"
THEN [Equipment ID] END) / COUNT([Equipment ID])

Purpose: Helps monitor the availability of hospital equipment and identify potential resource constraints.

Measurement : Percentage

3.2.9 Total Patient Transfers

Definition: Total number of patients who were transferred.

Formula: Total Patient Transfers = COUNTD(IF [Transfer To Department] <> "Not Transferred"
THEN [Patient ID] END)

Purpose: Measures the volume of patient movement between departments or hospital units.

Measurement : count

3.2.10 Transfer Rate

Definition: Percentage of patients who were transferred.

Formula: Transfer Rate = COUNTD(IF [Transfer To Department] <> "Not Transferred"
THEN [Patient ID] END) / COUNTD([Patient ID])

Purpose: Helps analyze patient movement and transfer patterns within the hospital.

Measurement : Percentage

3.2.11 Total No of Departements – 10

3.3 KPI Results

The calculated KPIs were integrated into Tableau visualizations and KPI cards.

The results can be dynamically updated based on dashboard filters such as:

- Year
- State
- Hospital
- Hospital Type
- Department
- Other dashboard-level selections

Therefore, KPI values may change when users interact with the dashboard.

4. Dashboard Development

4.1 Overview

The dashboard development stage focused on transforming the engineered healthcare KPIs into interactive and visually understandable dashboards using **Tableau**.

Four dashboards were developed to provide different perspectives of hospital operations:

1. Hospital Overview Dashboard
2. Patient Flow Dashboard
3. Department Analytics Dashboard
4. Resource Utilization Dashboard

The dashboards were designed with interactive visualizations, KPI cards, filters, and dashboard actions to help users explore hospital performance and identify important trends.

4.2 Technology Used

Technology	Purpose
Tableau Desktop	Dashboard development and visualization
Excel / CSV	Data source
Python Jupyter Notebook	Data cleaning and transformation
GitHub	Project and documentation management

4.3 Hospital Overview Dashboard

Purpose

The Hospital Overview Dashboard provides a high-level summary of hospital performance and allows users to quickly understand overall operational conditions.

Business Question	Answer	Chart Type
What is the total number of admissions?	10,000	KPI Card
What is the current bed occupancy rate?	36.8%	KPI Card
What is the readmission rate?	49.9%	KPI Card
Which admission type contributes most Admissions ?	Urgent Type	Donut Chart
How do admissions changing every month ?	The trend shows monthly increases or decreases in patient admissions.	Line Chart
How does occupancy changing every month ?	The trend shows monthly variations in hospital bed occupancy.	Line Chart
Is readmission rate increasing or decreasing over time ?	The trend shows monthly changes in the hospital readmission rate.	Line Chart
Which hospitals have the highest readmission percentage?	Deepak Hospital	Horizontal Bar Chart

4.4 Patient Flow Dashboard

Purpose

The Patient Flow Dashboard focuses on the movement of patients through the hospital, from admission to transfer and other stages of their hospital journey.

Business Question	Answer	Chart Type
What is the Average length Of Stay across the hospital?	78.78 days	KPI Card
Which months experience the highest and lowest patient admissions?	January -Highest November-Lowest	Bar chart
How have patient discharges changed over time?	The trend shows monthly changes in the patient discharges.	Area Chart
Which departments transfer the most patients ?	Neurology	Horizontal Bar Chart
What are the most common patient movement paths between departments ?	Nephrology to Oncology	Heat Map(matrix)

How has the average length of stay changed over time ?	The trend shows monthly variations in Average length of stay.	Line Chart
What is the transfer rate of patients?	17.6%	KPI Card
What are the total no of Patient Transfers ?	1760	KP Card

4.5 Department Analytics Dashboard

Purpose

The Department Analytics Dashboard evaluates the performance of individual hospital departments.

Business Question	Answer	Chart Type
What is the department efficiency score ?	29.15	KPI Card
Which departments have the highest and lowest Department Efficiency scores ?	ICU -Highest Nephrology -Lowest	Horizontal Bar chart
Which departments have the shortest and longest Average Length of Stay (ALOS) ?	Oncology -longest Cardiology-shortest	Horizontal Bar Chart
How does monthly patient volume vary across departments ?	The trend shows monthly variations in patient volumes across departments.	Line Chart
Which departments have the highest and lowest bed occupancy relative to their capacity ?	Psychiatry -Highest General Medicine- Lowest	Horizontal Bar Chart
Which department have the highest Readmission rate ?	Oncology	Horizontal Bar Chart
What is the transfer rate of patients?	17.6%	KPI Card
What are the total no of Departments?	10	KPI Card

4.6 Resource Utilization Dashboard

Purpose

The Resource Utilization Dashboard focuses on the utilization and availability of hospital resources such as beds, staff, and equipment.

Business Question	Answer	Chart Type
What is the bed Occupancy rate ?	37.3%	KPI Card

Which hospitals have the highest number of occupied beds ?	Medindia Hospital	Horizontal Bar chart
What is the Equipment Availability rate ?	32.9%	KPI Card
Which departments have the highest bed capacity ?	Neurology	Bar Chart
Which departments have the highest staff utilization percentage ?	Nephrology	Bar Chart
Which equipment types are used for the longest duration ?	Patient Monitor	Bar Chart
What is the distribution of Equipment status (Available , In use , Maintenance) ?	Available-32.9% Maintenance-33.6% In Use- 33.6%	Donut Chart
What is the Staff Utilization Rate ?	70.8%	KPI Card

5. Dashboard Integration

Interactive functionality was incorporated to make the dashboards dynamic and user-friendly.

The following features were implemented:

- **Filters** : Filters allow users to view data for specific years, states, hospitals, departments, and other relevant dimensions.
- **Dashboard Actions** : Dashboard actions were used to enable interaction between visualizations and filters.
- **Navigation Controls** : Navigation buttons were incorporated to allow users to move between the different dashboards.
- **Dashboard Linking** : The dashboards were connected through navigation controls, allowing users to move between different analytical perspectives.

6. Testing & Validation

6.1 Overview

The testing and validation stage was performed to ensure that the cleaned dataset, calculated KPIs, Tableau visualizations, filters, dashboard actions, and navigation controls worked correctly.

The objective was to verify the **accuracy, consistency, functionality, and usability** of the final healthcare analytics dashboard.

6.2 QA Checklist

Test Category	Status
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Data validation	Passed
KPI validation	Passed
Dashboard visualization testing	Passed
Dashboard-action filter testing	Passed
Navigation controls	Passed
Dashboard linking	Passed
Visual/layout validation	Passed

7. User Guide

User Workflow

Open Dashboard → Select Required Filters → Review KPIs → Interact with Visualizations → Navigate to Related Dashboard → Analyze Insights → Clear Filters for Overall View

8. Conclusion

The Hospital Analytics Dashboard project successfully transformed raw hospital data into a structured and interactive analytics solution. The project followed a systematic workflow covering data collection, data cleaning and transformation, healthcare KPI engineering, dashboard development, dashboard integration, testing, validation, and final documentation.

The hospital dataset was cleaned and prepared to improve data quality and consistency. Important healthcare KPIs such as **Total Admissions, Occupancy Rate, Average Length of Stay, Readmission Rate, Bed Utilization Rate, and Department Efficiency Score** were developed to measure hospital performance.

Four interactive Tableau dashboards were developed: **Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization**. These dashboards allow users to analyze hospital performance, patient movement, department-level operations, and resource utilization through interactive visualizations and filters.

Dashboard integration features such as **filters, navigation controls, dashboard linking, and parameter actions** improved the overall usability and interactivity of the solution. Testing and validation were also performed to ensure that KPIs, filters, visualizations, and dashboard actions function correctly.

Overall, the project demonstrates how data analytics and visualization can convert complex healthcare data into meaningful insights. The completed dashboard provides a user-friendly platform for monitoring hospital operations and supports **data-driven decision-making and performance analysis**.

9. References

Data sets: <https://www.kaggle.com/datasets/hassanjameelahmed/healthcare-analytics-patient-flow-data>

Github link : <https://github.com/springboardmentor09876x-cmd/Hospital-Operations-Patient-Analytics-/tree/Thota-Harini>